

AI Methods for Classification of Production Errors on Camera Images of Additive Manufacturing Processes

Ramya Subramanian Porselva Bharathi (224201309)

Universität Koblenz, 56070 Germany

Abstract

- Motivation: In additive manufacturing (AM), production errors such as porosity and layer shifting occur. Surface defects degrade quality and increase waste.
- Problem statement: How to automatically detect and classify these errors using camera images and AI methods.
- Approach: Review and analyze existing AI/computer vision approaches, and summarize the findings.
- Expected contributions: synthesis of methods, identification of gaps, and insights on improving quality, sustainability, and reliability in AM through AI-driven defect detection.

1 Introduction

1.1 Context & Importance

- Introduction of additive manufacturing and its increasing adoption in industries.
- Emphasize how defects reduce reliability, increase costs, and hinder adoption.

1.2 Problem Statement

- The core question: Which AI methods can reliably classify production errors from camera images in AM?
- Define “production errors” (Geometry, Thermal, Deposition, Surface, Process Errors) as the target classes.

1.3 Objectives

- Objective 1: Identify and evaluate state-of-the-art AI methods for camera-based defect classification in Additive Manufacturing.
- Objective 2: Compare their strengths, limitations, and suitability.

2 Theoretical Background

2.1 Additive Manufacturing Processes & Defects

- Describe major AM techniques for printing ceramics: FDM, SLM, SLS, DLP, etc.
- List common defects and how they manifest visually in images.

2.2 Camera-Based Monitoring in AM

- Explanation of imaging modalities: 2D Camera, Thermal/IR, Coaxial Imaging, 3D Imaging, etc.
- Challenges: lighting variation, reflections, shadows, motion blur, varying focus, depth of field, and sensor noise.
- Acquisition setups: fixed camera above print bed, side cameras, in-situ vs post-process imaging, etc.

2.3 AI Foundations Relevant to Classification

- Supervised learning, classification vs detection vs segmentation tasks, etc.
- Deep learning architectures
- Key processes: data preprocessing, feature extraction, training/validation split, and loss functions, etc.

3 Technical Background

3.1 Traditional Image Processing Approaches in AM Defect Detection

- Methods like thresholding, morphological operations, edge detection, texture descriptors, and their limitations.

3.2 Machine Learning Methods

- Feature extraction + classifiers such as SVM, Random Forest, kNN, etc.

3.3 Deep Learning Approaches

- End-to-end learning directly from images, using popular CNN or detection.

3.4 Comparative Analyses

- Summary table of methods

4 Methodology

4.1 Procedure for Searching Papers

- Search databases using keywords in additive manufacturing.

4.2 Procedure for Selecting Papers

- Select recent peer-reviewed papers focusing on AI methods in ceramic AM.

4.3 Comparison Criteria

- Compare methods based on AI technique, dataset used, performance metrics, material type, and identified challenges.

5 Results

5.1 Aggregate Performance Summary

- Comparative table to highlight which methods perform best under what conditions.
- Highlight trends

5.2 Example Results

- Consolidation of methods showing classified correctly and misclassified images.

5.3 Analysis of Strengths & Weaknesses

6 Discussion & Conclusions

6.1 Interpretation of Findings

- What do the results tell us about the feasibility and maturity of AI for camera-based defect classification in AM?
- Which architectures or strategies are promising?
- Impact on AM reliability, defect reduction, cost savings, and predictive maintenance.

6.2 Challenges & Limitations

- Data scarcity, domain shift between machines, real-time performance, explainability, and integration with AM control systems.

6.3 Future Work & Recommendations

References

1. Goodfellow, I., Bengio, Y., Courville, A.: Deep Learning. MIT Press, 2016.
2. Caggiano, A., Zhang, J., Alfieri, V., Caiazzo, F., Gao, R., Teti, R.: Machine learning-based image processing for online defect recognition in additive manufacturing. CIRP Annals 68(1), 451–454 (2019). <https://doi.org/10.1016/j.cirp.2019.03.021>

3. Wu, J., Huang, C., Li, Z., Li, R., Wang, G., Zhang, H.: An in situ surface defect detection method based on improved You Only Look Once algorithm for wire and arc additive manufacturing. *The International Journal of Advanced Manufacturing Technology* 123(5), 1609–1620 (2022).
<https://doi.org/10.1007/s00170-022-10602-8>
4. Saxena, S.K., Gopi, T., Nagesh, H., Gupta, M.K., Gaur, V., Krolczyk, G., Wu, C.: Machine learning techniques in additive manufacturing: a state of the art review on design, processes and production control. *J Intell Manuf* 34, 1–35 (2023).
<https://doi.org/10.1007/s10845-022-02029-5>
5. Chen, H.Y., Huang, Y.M., Hsu, Y.Y., Cheng, Y.S., & Lin, Y.S.: Deep Learning Applied to Defect Detection in Powder-Bed Fusion Additive Manufacturing. *Materials*, 15(16), 5662 (2022).
<https://doi.org/10.3390/ma15165662>
6. Eqbal, M.A., Eqbal, A., Khan, Z.A., Badruddin, I.A.: Machine Learning in Additive Manufacturing: A Comprehensive Insight. *International Journal of Lightweight Materials and Manufacture* (2024).
<https://doi.org/10.1016/j.ijlmm.2024.10.002>
7. Wang, W., Wang, P., Zhang, H., Chen, X., Wang, G., Lu, Y., Chen, M., Liu, H., Li, J.: A Real-Time Defect Detection Strategy for Additive Manufacturing Processes Based on Deep Learning and Machine Vision Technologies. *Micromachines* 15(1), 28 (2024).
<https://doi.org/10.3390/mi15010028>
8. Nikam, D., Chukwuemeka, A., Nigam, A., Bhosale, T., Nikam, S.H.: On the Application of YOLO-Based Object Detection Models to Classify and Detect Defects in Laser-Directed Energy Deposition Process. *Progress in Additive Manufacturing*, 10(10), 7609–7624 (2025).
<https://doi.org/10.1007/s40964-025-01056-x>
9. Deshpande, S., Venugopal, V., Kumar, M., et al.: Deep learning-based image segmentation for defect detection in additive manufacturing: an overview. *Int J Adv Manuf Technol* (2024).
<https://doi.org/10.1007/s00170-024-14191-6>
10. Cao, M., Fu, L., Ai, F., Zhou, K.: An Improved 3D Printing Extrusion Defect Detection Method Based on YOLO-v8. Preprint at Research Square (2024).
<https://doi.org/10.21203/rs.3.rs-3993733/v1>
11. Ruan, S., Zhan, C., Liu, B., Wan, Q., & Song, K.: A High Precision YOLO Model for Surface Defect Detection Based on PyConv and CISBA. *Scientific Reports*, 15(1), 91930 (2025).
<https://doi.org/10.1038/s41598-025-91930-z>
12. Wang, Z., Zhao, L., Li, H., Xue, X., Liu, H.: Research on a Metal Surface Defect Detection Algorithm Based on DSL-YOLO. *Sensors* 24(19), 6268 (2024).
<https://doi.org/10.3390/s24196268>